Econ 527 HW1

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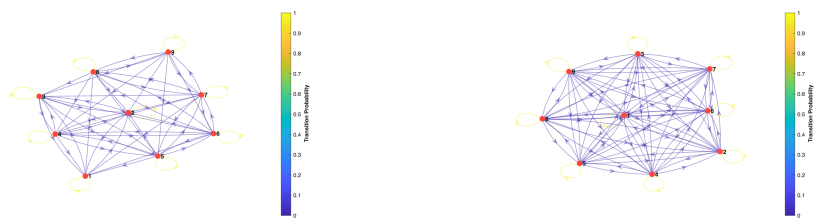
April 2026

1 Discretizing an AR(1)

We seek to approximate

$$x_t = \rho x_{t-1} + \epsilon \tag{1}$$
$$\epsilon \sim N(0, \sigma^2)$$

with parameters $\rho = 0.99$ and $\sigma = 0.2$.



(a) DTMC for Tauchen

(b) DTMC for Rouwenhorst

Figure 1: Generated Discrete Time Markov Chains

We use both Tauchen's method and Rouwenhorst's method to similaute 1000 histories (each 1000 periods long) from a grid with 9 points and 3 standard deviations.

Table 1: Comparison of AR(1) Discretization Methods

Moment	**Tauchen**	**Rouwenhorst**
First-order autocorrelation: $\text{cor}(x_t, x_{t-1})$	1.0077	0.9991
Unconditional variance: $\text{var}(x_t)$	0.4963	0.4921
First-order autocovariance: $\text{cov}(x_t, x_{t-1})$	0.5013	0.4970
Conditional variance: $\text{var}(x_t \mid x_{t-1}) = \text{var}(\varepsilon_t)$	24.9407	24.7273

In this case, Rouwenhorst is more accurate. From the comparison ratios in Table 1, the moments calculated from Rouwenhorst are smaller across the board.

Table 2: Comparison of Gridpoint Formation using Tauchen's Method

Moment	**3stdev**	**2.5stdevs**
First-order autocorrelation: $\text{cor}(x_t, x_{t-1})$	1.0077	1.0055
Unconditional variance: $\text{var}(x_t)$	0.4963	0.4952
First-order autocovariance: $\text{cov}(x_t, x_{t-1})$	0.5013	0.5002
Conditional variance: $\text{var}(x_t \mid x_{t-1}) = \text{var}(\varepsilon_t)$	24.9407	24.8860

Recalculating the same moments but changing the grid to have 2.5 standard deviations with Tauchen's method, we see that accuracy decreases in 2

2 Interpolation

We seek to interpolate a function using CRRA utility as our benchmark function. CRAA utility is defined as

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma} \tag{2}$$

with curvature parameter $\gamma = 4$.

1. An evenly spaced grid with consumption was created with 100 grid points on the interval $[0.1, 30]$ which was then used to create a natural spline interpolant.

   Using an evenly spaced grid with 1000 points, we generated the approximation $g_1(x)$, the results of which can be seen in Figure 2). The maximum error obained in this interval i.e. $\max |u(x_j) - g(x_j)| = 141.59$.

2. Because we have chosen CRAA utility, we know that $u(c)$ is monotonically increasing, striclty cincave and less than 0 on the interval $0.1, 30]$. However, it is not guaranteed that $g(x_j)$ has the same properties. From simple observation of Figure 2 we know the following are **not** true $\forall x$.

$$\begin{gathered} g_1(x_{j+1}) - g_1(x_j) > 0 \\ \frac{g_1(x_{j+2}) - g_1(x_{j+1})}{x_{j+2} - x_{j+1}} < \frac{g_1(x_{j+1}) - g_1(x_j)}{x_{j+1} - x_j} \\ g_1(x_j) < 0 \end{gathered}$$

   Since the interpolating function $g(x)$ not only doesn't preserve basic properties of $u(c)$ also also generates a maximum error of 144, it is in general not a good approximation.

3. To fix this issue, we investigate changing the grid form an evenly-spaced one to a polynomial grid over $[0.1, 30]$ with curvature parameter 3. We will define this new interpolative function $g_2(x)$. These were creates with using

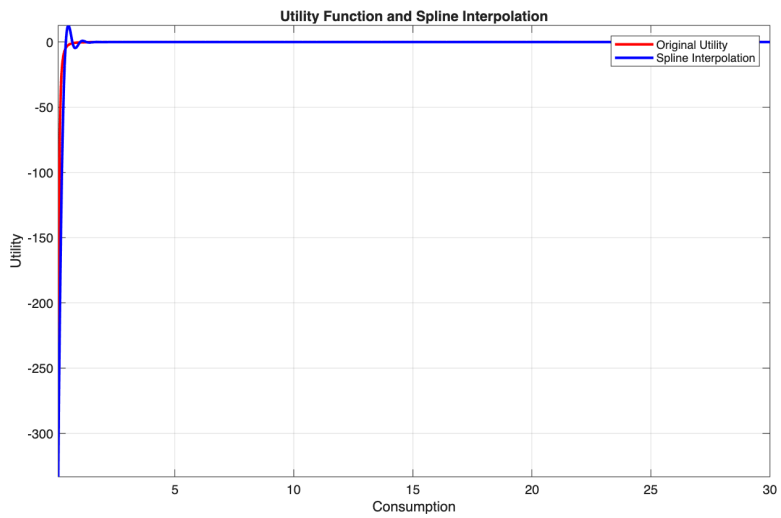


Figure 2: True function and Natural Spline Interpolation

$$x_j = 0.1 + (30 - 0.1)z_i^3$$

where z_1 is an evenly spaced grid with 1000 nodes over $[0, 1]$.

4. To then test the accuracy this model's extrapolation power, we plot x_j over 100 evenly spaced points on the interval $[30, 40]$ and $[00.1, 0.1]$ seen in Figure 3 and 4, respectively.

   In both cases, monotonicity, strict concavity, and $u(c) < 0$ are all preserved, i.e. the following are true

$$g_2(x_{j+1}) - g_2(x_j) > 0$$
$$\frac{g_2(x_{j+2}) - g_2(x_{j+1})}{x_{j+2} - x_{j+1}} < \frac{g_2(x_{j+1}) - g_2(x_j)}{x_{j+1} - x_j}$$
$$g_2(x_j) < 0$$

   on the defined intervals.

   Extrapolation on the interval $[30, 40]$ appears to be more accurate. The true function $u(c)$ is much more highly curved on $[0.01, 0.1]$ than on $[30, 40]$ and as such the interpolating function does not extrapolate well.

3 Description of code

- **PS_Q1.m** Matlab code for discretizing an AR(1) process

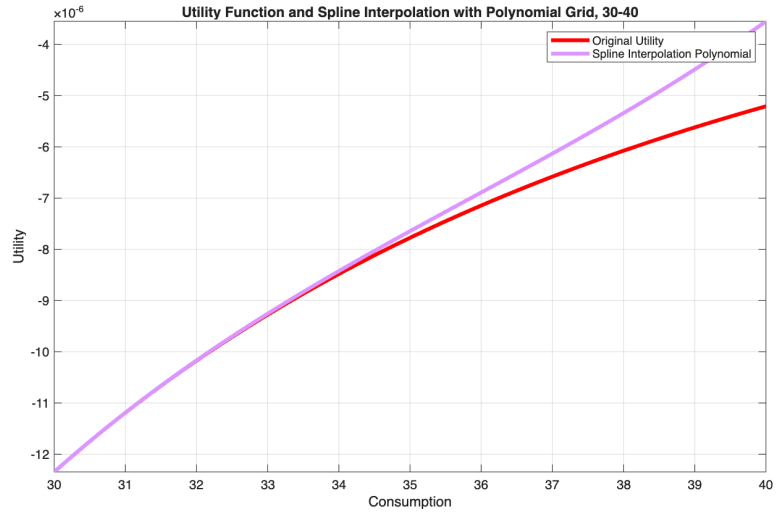


Figure 3: Extrapolation of $g_2(x)$ on $[30, 40]$

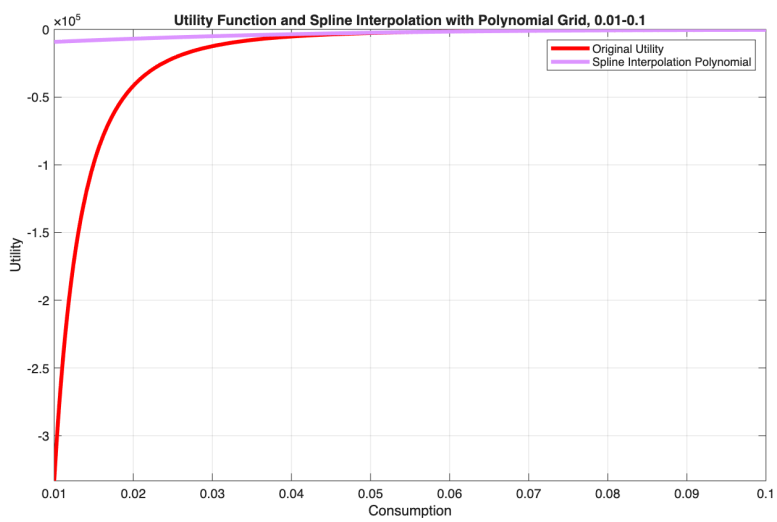


Figure 4: Extrapolation of $g_2(x)$ on $[0.01, 0.1]$

- **PS1_Q2.m** Matlab code for Inerpolation
- **tauchen.m** Matlab code for implementing Tauchen's method, provided by Professor Greaney
- **rouwenhorst.m** Matlab code for implementing Rouwenhorst's method, provided by Professor Greaney
- **mc_invdist.m** Matlab code for calcuating invariant distribution, provided by Professor Greaney